

Sheath Academy Implementation Plan - Features

20-23: Attendance Spine

Status

Finalized based on agreed product decisions.

Features Covered

- Feature 20 - Attendance record data model
- Feature 21 - Mark attendance by child/date
- Feature 22 - Optional hours/minutes field
- Feature 23 - Attendance summary

Purpose

This slice establishes the Wave 1C attendance spine. The app should move from only planning lessons to also recording whether school happened, optionally documenting time, and showing whether attendance records are complete for the selected child and school year.

The goal is record confidence without overbuilding legal/compliance logic.

Agreed MVP Decisions

Attendance status values

Use only:

```
type AttendanceStatus = "present" | "absent" | "partial";
```

Do not add these in this slice:

```
"excused" | "sick" | "holiday" | "not_recorded"
```

There is no stored `not_recorded` state. If no attendance record exists for a child/date, attendance has not been recorded.

Missing day rule

For MVP:

A missing attendance day is any Monday-Friday date inside the active school year, up to today, for the selected child, that has no `attendance_record`.

Weekend dates may still have attendance records, but weekends are not expected attendance days by default.

Do not account for holidays, breaks, custom school calendars, state-specific rules, or household-configurable school days yet.

Time tracking

Store optional time as total minutes:

```
minutes?: number;
```

The UI may display this as 2h 30m, but storage should remain integer minutes.

Quran and role implications

Program attendance, role permissions, and Quran/Hifz session attendance are valid future concerns. They are out of scope for this Wave 1C slice.

Do not treat Quran sessions as generic class attendance forever, but do not build the Quran-specific attendance model now.

Attendance and lesson completion

Attendance remains independent from lesson completion.

Completing lessons should not automatically mark attendance. Later, the app may suggest: “Lessons completed today; mark attendance?” but the parent should intentionally create the attendance record.

Feature 20 - Attendance Record Data Model

Goal

Create the canonical attendance data model.

Recommended module

```
src/features/attendance/
```

Suggested files:

```
src/features/attendance/types.ts
src/features/attendance/store.ts
src/features/attendance/validation.ts
src/features/attendance/utils/date.ts
src/features/attendance/utils/summary.ts
src/features/attendance/components/
```

TypeScript model

```
export type AttendanceStatus = "present" | "absent" | "partial";

export interface AttendanceRecord {
```

```

    id: string;
    childId: string;
    date: string; // YYYY-MM-DD, local calendar date
    status: AttendanceStatus;
    minutes?: number;
    notes?: string;
    createdAt: string;
    updatedAt: string;
  }

```

Required fields

Required at create/update time:

```

childId
date
status

```

Optional:

```

minutes
notes

```

System-generated:

```

id
createdAt
updatedAt

```

ID format

Use:

```
attendance_record_${Date.now()}_${counter}
```

Date format

Use local date-only strings:

```
YYYY-MM-DD
```

Do not store the attendance date as a full timestamp. A parent means “school happened on this date,” not a UTC instant.

Uniqueness rule

There should be only one attendance record per child per date.

Enforce:

```
childId + date must be unique
```

If a record already exists for the child/date, update it instead of creating a duplicate.

Validation

Validate:

```
childId exists
date is valid YYYY-MM-DD
status is present | absent | partial
minutes is optional
notes is optional plain text
```

Normalize optional strings:

```
notes: notes?.trim() || undefined
```

Done means

- AttendanceRecord type exists.
 - Attendance records can be listed, created, and updated.
 - `childId + date` uniqueness is enforced.
 - Status enum is `present | absent | partial`.
 - Missing attendance is represented by absence of a record, `not_recorded`.
 - Dates use YYYY-MM-DD.
-

Feature 21 - Mark Attendance by Child/Date

Goal

Parent can quickly mark attendance for a child/date and edit it later.

MVP UI pattern

Use quick-tap cards, not a full calendar grid.

Primary placement:

Today tab

Suggested UI:

Attendance Today

Adam

[Present] [Partial] [Absent]

Khadijah

[Present] [Partial] [Absent]

If one child is selected, show one child. If **All Children** is selected, show one row per child.

Behavior

Marking attendance should use upsert behavior.

```
upsertAttendanceRecord({  
  childId,  
  date,  
  status: "present"  
});
```

If a record exists, update it. If no record exists, create it.

Edit behavior

For MVP, editing can be simple: clicking a different status updates the existing record.

A fuller edit form can allow notes and minutes.

Optional bulk action

A **Mark all present** action is allowed only if it remains small and clear. It should create/update one record per child for the selected date.

This is useful, but not required for Feature 21 completion.

Do not build yet

- Bulk-mark a week
- Full attendance calendar grid
- Role-based attendance views
- Program/admin approval

Done means

- Parent can mark **present**, **absent**, or **partial** for child/date.
- Parent can update an existing record.
- Today tab supports selected child.
- Today tab supports **All Children** view.
- Records persist according to the app's current store behavior.

Feature 22 - Optional Hours/Minutes Field

Goal

Allow families who need time documentation to record it without burdening families who do not.

Data model

Use:

minutes?: **number**;

Validation

Recommended:

minutes optional
integer only
min 0
max 720

Blank is preferred over storing 0 unless the user explicitly enters it.

For absent records, clear minutes or leave blank.

Partial status behavior

partial does not require minutes, but the UI should gently show the optional time field.

Do not block saving a partial record just because minutes are blank.

Absent status behavior

Absent records may have notes.

Examples:

Sick day
Travel
Family emergency

Do not add separate **sick** or **excused** statuses yet.

Display format

Use:

2h 30m
45m

Utility:

```
formatAttendanceMinutes(minutes?: number): string
```

Examples:

```
undefined -> ""  
0 -> "0m"  
30 -> "30m"  
90 -> "1h 30m"  
120 -> "2h"
```

UI placement

Minutes should be unobtrusive.

Example:

```
Present [selected]  
Optional time: [ __ ] h [ __ ] m  
Notes: [ optional ]
```

A single minutes input is acceptable if faster to implement.

Done means

- Attendance record can store optional minutes.
 - Blank time is allowed.
 - Time is not required for all users.
 - Time displays in readable hours/minutes format.
 - Minutes are included in summary totals.
-

Feature 23 - Attendance Summary

Goal

Parent can see attendance count and missing days for the selected child and active school year.

Required scope

Feature 23 should support:

```
selected child  
active school year  
attendance counts  
missing day count
```

missing date list
total logged minutes

If All Children is selected, show per-child summaries. Do not merge all children into one ambiguous missing-days number.

MVP missing-day rule

Use this exact rule for now:

Expected attendance dates = Monday-Friday dates inside the active school year, from schoolYearStart to schoolYearEnd

A missing day is:

An expected attendance date for the selected child where no attendance_record exists.

Do not account for:

holidays
school breaks
custom homeschool schedules
field trip exceptions
state-specific rules

Summary type

```
export interface AttendanceSummary {  
  childId: string;  
  schoolYearId: string;  
  startDate: string;  
  endDate: string;  
  presentDays: number;  
  partialDays: number;  
  absentDays: number;  
  missingDays: number;  
  missingDates: string[];  
  totalMinutes: number;  
}
```

For All Children:

```
export interface AttendanceSummaryByChild {  
  summaries: AttendanceSummary[];  
}
```

Summary utility

Recommended functions:

```
getExpectedAttendanceDatesForSchoolYear(params: {  
  schoolYearStartDate: string;
```



```

    schoolYearEndDate: string;
    today: string;
  }): string[];

getAttendanceSummary(params: {
  childId: string;
  schoolYearId: string;
  records: AttendanceRecord[];
  schoolYearStartDate: string;
  schoolYearEndDate: string;
  today: string;
}): AttendanceSummary;

```

UI placement

Primary placement:

Records tab

Secondary lightweight indicator:

Today tab: Attendance marked / not marked today

Feature 23's fuller summary belongs in Records, because the parent is checking record completeness.

Summary copy

Example:

Attendance Summary - Adam

School year to date

Present: 42

Partial: 3

Absent: 2

Missing: 4

Logged time: 126h 30m

Missing attendance records:

Sep 12, Sep 19, Oct 3, Oct 10

Use careful wording:

Missing attendance records

Avoid overclaiming:

Non-compliant days

Invalid days

Legal issue

Done means

- Parent can see attendance summary for selected child/year.
 - Summary shows present, partial, absent, and missing counts.
 - Summary shows total logged minutes.
 - Summary lists missing dates.
 - Missing days follow the MVP Monday-Friday active-school-year rule.
 - All Children view shows per-child summaries.
-

Store/API Functions

Recommended attendance store functions:

```
listAttendanceRecords(): AttendanceRecord[];

getAttendanceByChildAndDate(
  childId: string,
  date: string
): AttendanceRecord | undefined;

upsertAttendanceRecord(input: UpsertAttendanceInput): AttendanceRecord;

deleteAttendanceRecord(id: string): void;

listAttendanceByChildAndDateRange(params: {
  childId: string;
  startDate: string;
  endDate: string;
}): AttendanceRecord[];
```

Input type:

```
export interface UpsertAttendanceInput {
  childId: string;
  date: string;
  status: AttendanceStatus;
  minutes?: number;
  notes?: string;
}
```

Components

Recommended components:

AttendanceQuickMarkCard
AttendanceStatusButtons
AttendanceTimeInput
AttendanceNotesInput
AttendanceSummaryCard
AttendanceMissingDatesList

Keep components small.

Do not build a full attendance management system yet.

Build Order

Step 1 - Feature 20

Build:

AttendanceRecord type
attendance store
validation
seed data
upsert behavior
child/date uniqueness

Step 2 - Feature 21

Build:

Today-tab quick mark card
status buttons
selected child behavior
All Children behavior
edit-by-upsert behavior

Step 3 - Feature 22

Build:

optional minutes input
minutes validation
minutes formatting helper
notes field if not already included

Step 4 - Feature 23

Build:

expected attendance date utility
school-year-to-date summary
missing-day calculation
summary card in Records tab
Today-tab marked/not-marked indicator if easy

Seed Data

Add light seed data only.

Recommended:

one present record for today
one partial record this week with minutes
one absent record this week
at least one missing weekday created by absence of a record

Do not flood seed data. The summary needs enough seed data to prove counts and missing dates work.

Manual Test Checklist

Feature 20

- Create an attendance record.
- Create/update a second record for the same child/date and confirm it updates instead of duplicating.
- Confirm `date` stores as YYYY-MM-DD.
- Confirm missing attendance is absence of record, not `not_recorded`.

Feature 21

- In Today tab, mark one child present.
- Change that child to partial.
- Change that child to absent.
- Switch child selector and confirm rows respect selected child.
- Select **All Children** and confirm one row per child.

Feature 22

- Save attendance with no minutes.

- Save partial attendance with minutes.
- Confirm minutes display as hours/minutes.
- Confirm absent clears or ignores minutes.
- Confirm notes can be stored for absent records.

Feature 23

- Open Records tab.
- Confirm selected child summary shows present, partial, absent, missing, and total minutes.
- Confirm missing dates are Monday-Friday only.
- Confirm missing dates stop at today, not future dates.
- Confirm All Children view shows per-child summaries.

Out of Scope

Do not implement in Features 20-23:

excused/sick/holiday statuses
state compliance logic
custom school calendars
holiday calendars
bulk week marking
PDF exports
report generation
records review checklist
role permissions
program/admin workflows
Quran-specific attendance sessions
AI nudges
alerts
automatic attendance from lesson completion

Implementation Prompt

Implement Features 20-23: Attendance Spine.

Create a feature-owned attendance module with **AttendanceRecord** as the canonical model. Use **childId**, **date**, **status**, optional **minutes**, optional **notes**, **createdAt**, and **updatedAt**. Store date as a local YYYY-MM-DD string. Use status values **present**, **absent**, and **partial**. Do not store **not_recorded**; missing attendance means no record exists. Enforce one attendance record per child per date with upsert behavior.

Add a Today-tab attendance quick-mark card. The parent can mark Present, Partial, or Absent for a selected child/date and edit by updating the existing record. If **All Children** is selected, show one row per child. A small **Mark all present** action is acceptable but not required.

Add optional time tracking using **minutes?: number**. Store total minutes as an integer. Display time as hours/minutes. Minutes are optional and should not burden users who do not track hours. Partial attendance should not require minutes. Absent records may include notes.

Add an attendance summary for the selected child and active school year. Use this MVP missing-day rule: Monday-Friday dates inside the active school year, up to today, require attendance; if no **attendance_record** exists for that child/date, count the date as missing. Weekends may have attendance records but are not expected by default. Show present, partial, absent, missing, total minutes, and missing dates. If **All Children** is selected, show per-child summaries.

Do not implement holidays, excused/sick statuses, state compliance logic, role permissions, Quran-specific attendance sessions, bulk week marking, reports/export, alerts, AI guidance, or automatic attendance from lesson completion yet.